

NBA Performance Analytics Dashboard

Overview

This project presents an interactive NBA analytics dashboard built using Power BI, designed to uncover insights into team performance, scoring trends, and roster composition. The dashboard enables dynamic exploration of season-wise and game-wise trends, team comparisons, and performance patterns through intuitive visualizations and filters.

Objectives

- Analyze **season-wise and game-wise scoring trends**
 - Compare **team performance across Eastern and Western conferences**
 - Derive **player insights using roster-based proxies**
 - Track **team rankings and win-loss patterns over time**
 - Enable **interactive filtering by season, team, and player**
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Key Features

Performance Trends

- Season-wise and game-wise analysis of **average points per game**
- Visualization of **league-wide scoring evolution over time**

Team & Conference Analysis

- Comparison of **team win percentages**
- Insights into **East vs West performance differences**

Player Insights (Proxy-Based)

- Analysis of **roster size and stability**
- Identification of teams with **largest and smallest rosters**
- Use of roster composition as a proxy for **player consistency and depth**

Rankings & Patterns

- Visualization of **win percentage trends over time**
- Snapshot-based **team rankings using win percentage**

Interactivity

- Dynamic filters for:
 - **Season**
 - **Team**
 - **Player**
 - Interactive exploration without disrupting long-term trend visuals
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KPIs Included

- Total Players (League)
 - Average Roster Size
 - Largest Team Roster
 - Smallest Team Roster
 - Roster Stability Score
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Tools & Technologies

- **Power BI** – Data visualization and dashboard creation
 - **DAX** – Measures and calculations
 - **Excel / CSV datasets** – Data preprocessing
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Key Insight

Due to the absence of player-level performance statistics, player insights were derived using **roster composition and continuity as proxy indicators**, reflecting real-world analytical adaptability.



How to Use

1. Open the Power BI dashboard file (.pbix)
2. Use the **filters (season, team, player)** to explore insights
3. Hover over visuals for detailed information

4. Analyze trends and comparisons interactively

Dashboard Preview

(Add screenshots here for better visibility)

Conclusion

This dashboard demonstrates how meaningful insights can be extracted even with limited data by applying structured analytical thinking, effective visualization techniques, and interactive design principles.
